

Passive expression pedal / I2C ADC / HAND ASSEMBLY

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = Into this board / OUT = out / I/O = both / GND = 0 V

J302 / PASSIVE PEDAL
1: TIP I/O 2: RING I/O
3: GND 0V

JP301 / PEDAL JUMPER
1: TIP 2: RING
3: PEDAL IN 4: EXC OUT
5: RING 6: TIP

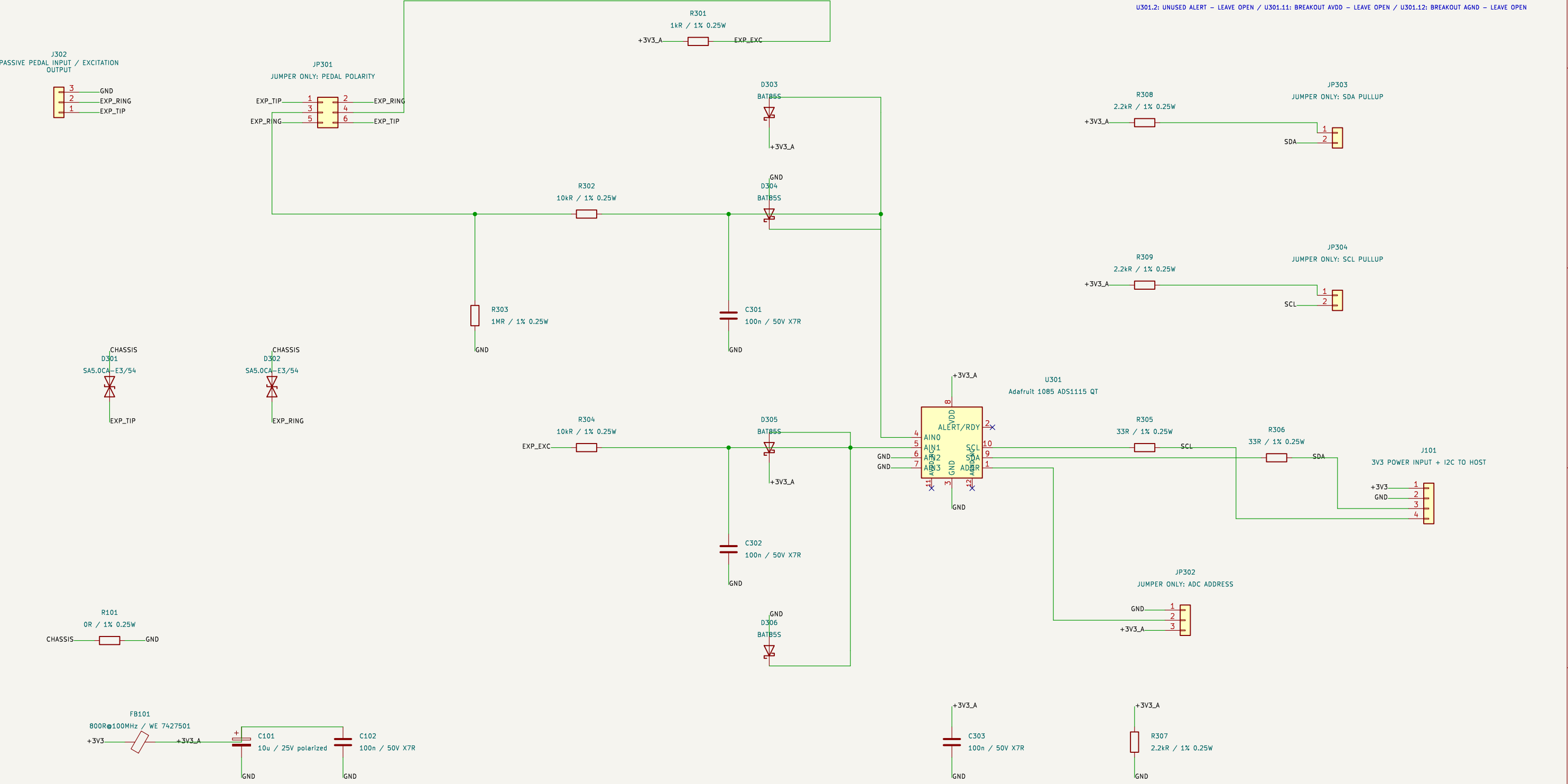
JP303 / DATA PULLUP
1: PULLUP 2: DATA I/O

J101 / PWR IN / I2C
1: 3V3 IN 2: GND 0V
3: DATA I/O 4: CLOCK IN

JP302 / ADDR JUMPER
1: LOW GND 2: SELECT
3: HIGH 3V3

JP304 / CLOCK PULLUP
1: PULLUP 2: CLOCK IN

U301.2: UNUSED ALERT - LEAVE OPEN / U301.11: BREAKOUT AVDD - LEAVE OPEN / U301.12: BREAKOUT AGND - LEAVE OPEN



Passive 10k–100k pot only. Sleeve = GND. JP301: both shunts 1–3 / 2–4 for tip wiper; 3–5 / 4–6 for ring wiper.
JP302: 1–2 = 0x48; 2–3 = 0x49. Never leave ADDR floating. JP303/304 enable local 2.2k I2C pull-ups.
Remove BOTH pull-up shunts when host already has pull-ups (Raspberry Pi). Start at 100 kHz I2C.
ADS1115: +/-4.096 V, 128 SPS. Read AIN0 and AIN1; normalize AIN0/AIN1, calibrate endpoints and smooth.

Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Expression_THT.kicad_sch

Title: Passive expression pedal / I2C ADC / HAND ASSEMBLY

Size: A3 Date: 2026–10–08

Rev: T1

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